



Irrigation and Nitrogen Schedule for Enhancing Nutrient Uptake, Grain Nutritional Quality and Protein Content of Wheat (*Triticum aestivum* L.) under Semi-arid Conditions

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ABSTRACT

Background: Efficient irrigation and nitrogen management are essential for improving nutrient use efficiency and grain nutritional quality of wheat (*Triticum aestivum* L.) under semi-arid conditions. Synchronizing irrigation with crop water demand and nitrogen application with critical growth stages can enhance nutrient uptake, protein synthesis and fertilizer use efficiency. The present study was conducted to evaluate the influence of different irrigation regimes and nitrogen scheduling practices on nutrient content, nutrient uptake and protein content of wheat.

Methods: A field experiment was conducted during the *Rabi* season of 2024-25 at the Research Farm of Vivekananda Global University, Jaipur, Rajasthan, India. The experiment was laid out in a split-plot design with three replications, comprising three irrigation regimes [I_1 (three irrigations), I_2 (four irrigations) and I_3 (five irrigations)] and three nitrogen scheduling treatments [N_1 (50% basal + 50% at crown root initiation), N_2 (50% basal + 25% at crown root initiation + 25% at jointing) and N_3 (33.33% each at basal, crown root initiation and jointing)]. Nitrogen, phosphorus and potassium content and uptake in grain and straw and grain protein content were determined using standard analytical methods and the data were analyzed using analysis of variance.

Result: Irrigation regimes and nitrogen scheduling significantly influenced nutrient content, uptake and grain protein content, whereas their interaction remained non-significant. The five-irrigation (I_3) recorded the highest nitrogen (1.693%), phosphorus (0.369%) and potassium (0.485%) contents in grain, together with maximum nitrogen (75.63 kg ha⁻¹), phosphorus (16.53 kg ha⁻¹) and potassium (21.72 kg ha⁻¹) uptake by grain and the highest and protein content (10.58%). Among the N levels tested nitrogen applied in three equal splits at basal, crown root initiation and jointing stages (N_3) significantly enhanced nutrient concentration and grain protein content, recording the highest grain nitrogen (1.694%), phosphorus (0.370%), potassium (0.484%) and protein content (10.59%). Under the conditions of the present study, five irrigations (I_3) recorded the highest N, P and K contents and uptake in grain and straw, while three equal split applications of nitrogen (N_3) resulted in the highest nutrient contents, nutrient uptake and grain protein content indicate that integrated management of irrigation and split nitrogen application effectively improves nutrient use efficiency and grain nutritional quality of wheat under semi-arid conditions.

Key words: Grain protein content, Irrigation regimes, Nitrogen scheduling, Nutrient uptake, Nutrient use efficiency.

INTRODUCTION

Wheat (*Triticum aestivum* L.) is one of the most important cereal crops cultivated worldwide and serves as a major source of food and nutritional security. Belonging to the family Poaceae, bread wheat is an allohexaploid species ($2n = 6x = 42$; AABBDD) possessing wide genetic adaptability to diverse agro-climatic conditions. It contributes nearly 20% of the calories and protein consumed by the global population and is valued for its high carbohydrate content, moderate-quality protein (10-14%), dietary fibre, B-complex vitamins and essential minerals such as iron and zinc (Singh, 2023; Reddy *et al.*, 2025). Besides serving as the raw material for bread, chapati, pasta, biscuits and numerous bakery products, wheat straw is an important component of livestock feeding systems and is widely utilized as organic manure and mulch in agricultural production systems.

Globally, wheat is cultivated over approximately 220 million hectares, producing more than 790 million tonnes annually, making it one of the world's leading cereal crops

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(FAO, 2024). India is the second-largest producer of wheat after China, with about 30.36 million hectares under cultivation and a production of 112.74 million tonnes during

the 2023-24 *rabi* season, recording an average productivity of 3.71 t ha⁻¹ (Ministry of Agriculture and Farmers Welfare, 2024). Rajasthan contributes substantially to national wheat production, cultivating nearly 2.92 million hectares with a production of approximately 11.3 million tonnes and an average productivity of 3.87 t ha⁻¹. However, wheat cultivation in the semi-arid regions of Rajasthan continues to face challenges due to erratic rainfall, declining groundwater resources and inefficient management of irrigation and fertilizers, which adversely affect nutrient use efficiency and grain quality (Zhao *et al.*, 2025).

Among the essential plant nutrients, nitrogen is considered the most critical for wheat production because it forms an integral component of amino acids, proteins, nucleic acids, enzymes and chlorophyll (Santhoshini *et al.*, 2023). Adequate nitrogen availability enhances leaf area development, photosynthesis, biomass accumulation and grain filling, whereas nitrogen deficiency reduces nutrient assimilation, protein synthesis and ultimately grain quality (Tahir *et al.*, 2024). Since, grain protein content is closely associated with nitrogen accumulation during grain filling, improving nitrogen uptake and utilization has become a major objective of sustainable wheat production (Nair *et al.*, 2026). Nevertheless, nitrogen use efficiency (NUE) in conventional wheat production systems generally remains between 30 and 40%, mainly because of volatilization, denitrification and leaching losses resulting from improper fertilizer management (Yadav *et al.*, 2023; Lone *et al.*, 2025).

Efficient irrigation management plays an equally important role in determining nutrient uptake and utilization by maintaining favorable soil moisture conditions for nutrient dissolution, diffusion and mass flow towards plant roots (Rimmi *et al.*, 2023; Chouhan *et al.*, 2023). Adequate soil moisture enhances root proliferation, nutrient absorption and translocation, thereby increasing the accumulation of nitrogen, phosphorus and potassium in grain and straw. In contrast, moisture stress during critical stages such as crown root initiation (CRI), tillering, jointing, flowering and grain filling restricts nutrient uptake, reduces nutrient use efficiency and lowers grain protein content (Tyagi *et al.*, 2022; Reddy *et al.*, 2023). Therefore, precise irrigation scheduling is essential not only for improving crop productivity but also for maximizing fertilizer use efficiency.

Recent studies have demonstrated that synchronizing nitrogen application with crop demand through split application substantially improves nutrient uptake, nutrient use efficiency and grain nutritional quality compared with conventional fertilizer practices (Gill *et al.*, 2019; Narolia *et al.*, 2016). Split nitrogen application at basal, crown root initiation and jointing stages maintains a continuous supply of available nitrogen throughout crop growth, resulting in enhanced root development, higher nutrient absorption and greater accumulation of nitrogen in developing grains (Kumar and Tripathi, 2021). Moreover, integrated management of irrigation and nitrogen has been reported to improve the uptake of phosphorus and potassium by

promoting balanced plant nutrition and efficient source-sink relationships, thereby increasing protein content and overall nutrient use efficiency (Verma *et al.*, 2021; Pradhan *et al.*, 2014).

Improving nutrient uptake and nutrient utilization has become increasingly important under changing climatic conditions, particularly in semi-arid regions where water scarcity and declining soil fertility often limit crop productivity. Integrated water and nitrogen management not only enhances nutrient acquisition and grain nutritional quality but also reduces nutrient losses and environmental pollution associated with inefficient fertilizer use (Ali *et al.*, 2021; Prem *et al.*, 2024). Therefore, the present investigation was undertaken to evaluate the influence of different irrigation regimes and nitrogen scheduling practices on nitrogen, phosphorus and potassium content, nutrient uptake and protein content of wheat under semi-arid conditions.

MATERIALS AND METHODS

Experimental site and climatic conditions

A field experiment was conducted during the *Rabi* season of 2024-25 at the Research Farm of Vivekananda Global University, Jaipur, Rajasthan, India, to evaluate the influence of irrigation regimes and nitrogen scheduling on nutrient content, uptake and protein content of wheat. The experimental site is situated at 26°05' N latitude and 75°28' E longitude, with an altitude of 427 m above mean sea level and falls under Agro-climatic Zone IIIA (Semi-Arid Eastern Plains) of Rajasthan.

The experimental region experiences a semi-arid climate characterized by hot summers and cool winters. During the crop-growing season, the maximum and minimum temperatures ranged from 37.9 to 17.3°C and 21.5 to 0.3°C, respectively. A total rainfall of 94.5 mm was received during the cropping period, with rainfall occurring mainly during the monsoon months, while the wheat crop was grown under irrigated conditions.

Soil characteristics

Before sowing, composite soil samples were collected from the 0-15 cm soil depth from different locations within the experimental field to determine the initial physico-chemical properties of the soil. The collected samples were air-dried, ground and passed through a 2-mm sieve before analysis following standard laboratory procedures.

Experimental design and treatments

The experiment consisted of nine treatment combinations comprising three irrigation regimes and three nitrogen scheduling treatments, arranged in a split-plot design with three replications. Irrigation regimes were assigned to the main plots, whereas nitrogen treatments were allotted to the sub-plots using the randomization.

The irrigation treatments were imposed according to the crop growth stages and dates recorded during the

experiment. Irrigation was applied through the check-basin method. The first, second, third, fourth and fifth irrigations were applied on 09 December 2024, 31 December 2024, 13 January 2025, 04 February 2025 and 06 March 2025, respectively, corresponding approximately to the crown root initiation (CRI), late tillering, jointing, flowering and grain-filling stages of wheat. The I_1 treatment received the first three irrigations, I_2 received the first four irrigations and I_3 received all five irrigations.

The irrigation treatments were:

I_1 : Three irrigations at CRI, late tillering and jointing stages.

I_2 : Four irrigations at CRI, late tillering, jointing and flowering stages.

I_3 : Five irrigations at CRI, late tillering, jointing, flowering and grain-filling stages.

The nitrogen scheduling treatments were:

N_1 : 50% nitrogen at basal + 50% at crown root initiation (CRI).

N_2 : 50% nitrogen at basal + 25% at CRI + 25% at jointing stage.

N_3 : 33.33% nitrogen each at basal, CRI and jointing stage

The experiment was conducted using wheat with a seed rate of 100 kg ha⁻¹. Each gross plot measured 3.0 × 5.0 m (15 m²), while the net plot size was 2.55 × 5.0 m (12.75 m²). Wheat was sown at a spacing of 22.5 cm between rows.

Crop management

The experimental field was prepared by one deep ploughing followed by repeated harrowing and planking to obtain a fine tilth. Wheat was sown during the *Rabi* season using recommended agronomic practices. Irrigation was applied according to the respective treatment schedules through the check basin method. Nitrogen fertilizer was supplied through urea, while phosphorus and potassium were applied through diammonium phosphate (DAP) and muriate of potash (MOP), respectively. The recommended fertilizer dose consisted of 120 kg N, 60 kg P₂O₅ and 40 kg K₂O ha⁻¹. Entire phosphorus and potassium were applied as basal, whereas nitrogen was applied according to the respective nitrogen scheduling treatments.

Determination of nutrient content

Representative grain and straw samples collected after harvest were air-dried, ground and analysed for nutrient concentration. Nitrogen concentration (%) was determined using the colorimetric method after sulphuric acid digestion with hydrogen peroxide, employing Nessler's reagent for colour development as described by Snell (1949). Phosphorus concentration (%) was estimated by the vanadomolybdophosphoric yellow colour method following tri-acid digestion according to Jackson (1973). Potassium concentration (%) was determined from the same digest using a flame photometer following the procedure described by Jackson (1973).

Protein content

Protein analysis of grain was calculated by multiplying N content (%) by 6.25 (based on assumption that N content 16 % of protein).

$$\text{Protein content (\%)} = \text{N content in \%} \times 6.25$$

Protein yield (kg ha⁻¹) =

$$\frac{\text{Protein content (\%)} \times \text{Grain yield (kg/ha)}}{100}$$

Chemical analysis

NPK content in grain and straw

NPK uptake in grain (kg ha⁻¹) =

$$\frac{\text{NPK content in grain (\%)} \times \text{Grain yield (kg/ha)}}{100}$$

NPK uptake in straw (kg ha⁻¹) =

$$\frac{\text{NPK content in straw (\%)} \times \text{Straw yield (kg/ha)}}{100}$$

Statistical analysis

The experimental data were subjected to analysis of variance (ANOVA) appropriate for a split-plot design following the procedure described by Panse and Sukhatme (1985). The significance of treatment effects was tested using the F-test at the 5% probability level. Wherever treatment effects were significant, critical difference (CD) at P = 0.05 was calculated for comparison of treatment means. Standard error of mean (SEm±) and coefficient of variation (CV) were computed for all measured parameters.

RESULTS AND DISCUSSION

Analysis of variance for nutrient content, nutrient uptake and protein content

The analysis of variance revealed that irrigation regimes and nitrogen scheduling significantly influenced nutrient content, uptake and protein content of wheat, whereas their interaction (I × N) was non-significant for all the parameters studied (Table 1a and 1b). Replication effects were also found to be non-significant for all the observed traits. The analysis further indicated that irrigation regimes exerted a significant effect on nitrogen, phosphorus and potassium contents in both grain and straw, as well as on protein content. Likewise, nitrogen scheduling significantly influenced all nutrient content parameters and protein content. However, the interaction between irrigation regimes and nitrogen scheduling did not show any significant effect, indicating that both factors acted independently in determining nutrient concentration and grain protein content (Table 1a). Similarly, irrigation regimes and nitrogen scheduling significantly affected the uptake of nitrogen, phosphorus and potassium by both grain and straw. In contrast, the interaction between irrigation regimes and nitrogen scheduling remained non-significant for all nutrient uptake parameters, suggesting that the response of nutrient uptake to irrigation regimes was consistent across

Table 1a: Analysis of variance (mean sum of squares) for nutrient content and protein content of wheat.

Source of variation	d.f.	N content (%) grain	N content (%) straw	P content (%) grain	P content (%) straw	K content (%) grain	K content (%) straw	Protein content (%)
Replication (R)	2	0.0018	0.0007	0.0004	0.000039	0.0001	0.0046	0.0374
Irrigation regimes (I)	2	0.0114**	0.0031**	0.0052**	0.000232**	0.0055**	0.0941**	0.4439**
Error (a)	4	0.0008	0.0002	0.0003	0.000010	0.0002	0.0053	0.0896
Nitrogen scheduling (N)	2	0.0127**	0.0028**	0.0052**	0.000147**	0.0051**	0.0955**	0.4968**
I × N	4	0.000002	0.00000	0.00001	0.000000	0.00000	0.00013	0.00008
Error (b)	12	0.0005	0.0001	0.0001	0.000004	0.0001	0.0045	0.0393

Note: Significant at $P \leq 0.05$.**Table 1b:** Analysis of variance (mean sum of squares) for nutrient uptake of wheat.

Source of variation	d.f.	N uptake (kg ha ⁻¹) grain	N uptake (kg ha ⁻¹) straw	P uptake (kg ha ⁻¹) grain	P uptake (kg ha ⁻¹) straw	K uptake (kg ha ⁻¹) grain	K uptake (kg ha ⁻¹) straw
Replication (R)	2	10.2520	3.2061	1.2720	0.1280	2.9229	69.4522
Irrigation regimes (I)	2	493.1645**	60.2173**	44.5499**	4.1261**	62.7595**	1100.5637**
Error (a)	4	36.0830	3.6832	1.5791	0.2326	3.6345	69.0512
Nitrogen scheduling (N)	2	525.0611**	56.2282**	46.2604**	3.3454**	62.5354**	1081.6652**
I × N	4	2.08230	0.14768	0.37566	0.00922	0.39926	4.77582
Error (b)	12	17.5181	2.8191	1.0405	0.0819	1.6037	63.3198

Note: Significant at $P \leq 0.05$.

different nitrogen scheduling treatments and vice versa (Table 1b).

Effect of irrigation regimes and nitrogen scheduling on nutrient content and protein content

Nitrogen content

Irrigation regimes and nitrogen scheduling significantly influenced N content in both grain and straw (Table 2). I_3 (five irrigations) recorded the highest N content (1.693% in grain and 0.497% in straw), compared with the lowest values under I_1 (1.622 and 0.460%). The corresponding increases under I_3 over I_1 and I_2 were 4.38 and 2.05% in grain and 8.04 and 3.97% in straw, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest N content (1.694% in grain and 0.496% in straw), whereas N_1 (50% N at basal + 50% at CRI) recorded the lowest (1.619 and 0.461%). Relative to N_1 and N_2 , N_3 increased N content by 4.63 and 1.99% in grain and 7.59 and 3.55% in straw, respectively.

Phosphorus content

Irrigation regimes and nitrogen scheduling significantly influenced P content in grain and straw (Table 2). I_3 (five irrigations) recorded the highest P content (0.369% in grain and 0.127% in straw), while the lowest values were observed under I_1 (0.321 and 0.117%). The increases under I_3 over I_1 and I_2 were 14.95 and 5.73% in grain and 8.55 and 3.25% in straw, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest P content (0.370% in grain and 0.126% in straw), whereas N_1 (50% N at basal + 50% at CRI) recorded the lowest (0.322 and 0.118%).

Compared with N_1 and N_2 , N_3 increased P content by 14.91 and 6.63% in grain and 6.78 and 2.44% in straw, respectively.

Potassium content

Irrigation regimes and nitrogen scheduling significantly influenced K content in grain and straw (Table 2). I_3 (five irrigations) recorded the highest K content (0.485% in grain and 1.691% in straw), while the lowest values were observed under I_1 (0.437 and 1.498%). The increases under I_3 over I_1 and I_2 were 10.98 and 2.97% in grain and 12.88 and 9.95% in straw, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest K content (0.484% in grain and 1.692% in straw), whereas N_1 (50% N at basal + 50% at CRI) recorded the lowest (0.438 and 1.496%). Compared with N_1 and N_2 , N_3 increased K content by 10.50 and 2.54% in grain and 13.10 and 9.94% in straw, respectively.

Protein content

Irrigation regimes and nitrogen scheduling significantly influenced grain protein content (Table 2). I_3 (five irrigations) recorded the highest protein content (10.58%), compared with the lowest value under I_1 (10.14%); the increase under I_3 over I_1 and I_2 was 4.34 and 2.02%, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest protein content (10.59%), whereas N_1 (50% N at basal + 50% at CRI) recorded the lowest (10.12%). The increase under N_3 over N_1 and N_2 was 4.64 and 2.02%, respectively.

Table 2: Effect of irrigation regimes and nitrogen scheduling on nitrogen, phosphorus and potassium content in grain and straw of wheat.

Treatments	Nitrogen content (%)		Phosphorus content (%)		Potassium content (%)		Protein content (%)
	Grain	Straw	Grain	Straw	Grain	Straw	
Irrigation regimes							
I ₁ - 3 irrigations	1.622	0.460	0.321	0.117	0.437	1.498	10.14
I ₂ - 4 irrigations	1.659	0.478	0.349	0.123	0.471	1.538	10.37
I ₃ - 5 irrigations	1.693	0.497	0.369	0.127	0.485	1.691	10.58
SEm±	0.009	0.004	0.006	0.001	0.005	0.024	0.10
CD (P = 0.05)	0.036	0.016	0.024	0.004	0.019	0.095	0.39
CV (%)	1.67	2.62	5.34	2.58	3.15	4.62	2.89
Nitrogen scheduling							
N ₁ - 50% basal + 50% at CRI	1.619	0.461	0.322	0.118	0.438	1.496	10.12
N ₂ - 50% basal + 25% at CRI + 25% at jointing stage	1.661	0.479	0.347	0.123	0.472	1.539	10.38
N ₃ - 33.33% basal + 33.33% at CRI + 33.33% at jointing stage	1.694	0.496	0.370	0.126	0.484	1.692	10.59
SEm±	0.008	0.003	0.003	0.001	0.003	0.022	0.07
CD (P = 0.05)	0.024	0.010	0.010	0.002	0.008	0.069	0.20
CV (%)	1.40	1.94	2.68	1.70	1.74	4.27	1.91

Note: CRI = Crown root initiation.

Effect of irrigation regimes and nitrogen scheduling on nutrient uptake**Nitrogen uptake**

Irrigation regimes and nitrogen scheduling significantly influenced nitrogen uptake by grain and straw (Table 3). I_3 (five irrigations) recorded the highest N uptake (75.63 kg ha⁻¹ in grain and 31.83 kg ha⁻¹ in straw), whereas I_1 (three irrigations) recorded the lowest (60.88 and 26.67 kg ha⁻¹, respectively). The increases under I_3 over I_1 and I_2 were 24.23 and 12.75% in grain and 19.35 and 10.02% in straw, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest N uptake (75.92 kg ha⁻¹ in grain and 31.66 kg ha⁻¹ in straw), while N_1 (50% N at basal + 50% at CRI) recorded the lowest (60.72 and 26.66 kg ha⁻¹, respectively). Compared with N_1 and N_2 , N_3 increased N uptake by 25.03 and 13.40% in grain and 18.76 and 8.76% in straw, respectively.

Phosphorus uptake

Irrigation regimes and nitrogen scheduling significantly influenced phosphorus uptake by grain and straw (Table 3). I_3 (five irrigations) recorded the highest P uptake (16.53 kg ha⁻¹ in grain and 8.13 kg ha⁻¹ in straw), whereas I_1 (three irrigations) recorded the lowest (12.08 and 6.77 kg ha⁻¹, respectively). The increases under I_3 over I_1 and I_2 were 36.84 and 16.82% in grain and 20.09 and 9.27% in straw, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest P uptake (16.63 kg ha⁻¹ in grain and 8.04 kg ha⁻¹ in straw), while N_1 (50% N at basal + 50% at CRI) recorded the lowest (12.11 and 6.82 kg ha⁻¹, respectively). Compared with N_1

and N_2 , N_3 increased P uptake by 37.32 and 18.53% in grain and 17.89 and 7.63% in straw, respectively.

Potassium uptake

Irrigation regimes and nitrogen scheduling significantly influenced potassium uptake by grain and straw (Table 3). I_3 (five irrigations) recorded the highest K uptake (21.72 kg ha⁻¹ in grain and 108.33 kg ha⁻¹ in straw), whereas I_1 (three irrigations) recorded the lowest (16.44 and 86.84 kg ha⁻¹, respectively). The increases under I_3 over I_1 and I_2 were 32.12 and 13.78% in grain and 24.75 and 16.39% in straw, respectively. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest K uptake (21.73 kg ha⁻¹ in grain and 108.08 kg ha⁻¹ in straw), while N_1 (50% N at basal + 50% at CRI) recorded the lowest (16.46 and 86.58 kg ha⁻¹, respectively). Compared with N_1 and N_2 , N_3 increased K uptake by 32.02 and 14.01% in grain and 24.83 and 15.47% in straw, respectively.

Effect of irrigation regimes on nutrient content, uptake and protein content

Adequate irrigation significantly improved the nutrient composition and nutritional quality of wheat. The highest nitrogen, phosphorus and potassium contents in both grain and straw, along with maximum protein content, were recorded under the five-irrigation treatment (I_3). Improved soil moisture under optimum irrigation enhances nutrient solubility, diffusion and root absorption, resulting in greater nutrient translocation from vegetative tissues to developing grains. Similar improvements in nutrient concentration under adequate irrigation have been reported by Patidar and Mali (2004) and Verma *et al.* (2021).

Table 3: Effect of irrigation regimes and nitrogen scheduling on nitrogen, phosphorus and potassium uptake by grain and straw of wheat.

Treatments	Nitrogen uptake (kg ha ⁻¹)		Phosphorus uptake (kg ha ⁻¹)		Potassium uptake (kg ha ⁻¹)	
	Grain	Straw	Grain	Straw	Grain	Straw
Irrigation regimes						
I_1 - 3 irrigations	60.88	26.67	12.08	6.77	16.44	86.84
I_2 - 4 irrigations	67.08	28.93	14.15	7.44	19.09	93.08
I_3 - 5 irrigations	75.63	31.83	16.53	8.13	21.72	108.33
SEm±	2.00	0.64	0.42	0.16	0.64	2.77
CD (P = 0.05)	7.86	2.51	1.64	0.63	2.50	10.88
CV (%)	8.85	6.59	8.82	6.48	9.99	8.65
Nitrogen scheduling						
N_1 - 50% basal + 50% at CRI	60.72	26.66	12.11	6.82	16.46	86.58
N_2 - 50% basal + 25% at CRI + 25% at jointing stage	66.95	29.11	14.03	7.47	19.06	93.60
N_3 - 33.33% basal + 33.33% at CRI + 33.33% at jointing stage	75.92	31.66	16.63	8.04	21.73	108.08
SEm±	1.40	0.56	0.34	0.10	0.42	2.65
CD (P = 0.05)	4.30	1.72	1.05	0.29	1.30	8.17
CV (%)	6.17	5.76	7.16	3.84	6.64	8.28

Note: CRI = Crown root initiation.

Significantly higher uptake of nitrogen, phosphorus and potassium under I_3 was primarily associated with increased grain and straw yields together with improved nutrient concentration. Since nutrient uptake is a function of nutrient content and biomass production, enhanced crop growth under adequate irrigation resulted in greater accumulation of nutrients in both grain and straw. Similar findings were reported by Parihar and Tiwari (2003), who observed increased nutrient uptake with improved irrigation scheduling. Yadav and Vyas (2006) also reported that optimum irrigation enhanced nutrient absorption through improved root activity and greater soil moisture availability. Likewise, (Kumar *et al.*, 2018; Ahmad and Kumar, 2015) concluded that maintaining favourable soil moisture significantly increased nutrient uptake and improved nutrient use efficiency in wheat. The higher protein content recorded under I_3 can be attributed to increased nitrogen accumulation in grains, as grain protein synthesis is closely associated with nitrogen assimilation during grain filling.

Effect of nitrogen scheduling on nutrient content, nutrient uptake and protein content

Nitrogen scheduling significantly enhanced nutrient concentration, nutrient uptake and protein content of wheat. Application of nitrogen in three equal splits at basal, crown root initiation (CRI) and jointing stages (N_3) consistently produced the highest nitrogen, phosphorus and potassium contents in grain and straw, as well as maximum nutrient uptake and grain protein content. The superiority of N_3 may be attributed to continuous nitrogen availability throughout the crop growth period, which promoted root development, enhanced nutrient absorption and improved assimilate production. Split nitrogen application synchronized nutrient supply with crop demand, thereby improving nitrogen use efficiency and facilitating greater uptake of phosphorus and potassium. Similar observations were reported by Meena *et al.* (2018); Kumar *et al.* (2018) who observed significantly higher nutrient uptake under balanced nutrient management. Singh and Singh (2017) also reported that improved nitrogen availability enhanced nutrient translocation and accumulation in wheat grains.

The increased uptake of nitrogen, phosphorus and potassium under N_3 resulted from the combined effect of higher nutrient concentration and greater biological yield. Continuous nutrient availability maintained active root growth and improved absorption from the rhizosphere, ultimately enhancing nutrient accumulation in both grain and straw. Similar findings have been reported by Parewa *et al.* (2021) and Kumari *et al.* (2022), who concluded that split nitrogen application significantly improved nutrient uptake, grain quality and nutrient use efficiency in wheat.

CONCLUSION

The present study demonstrated that irrigation regimes and nitrogen scheduling significantly influenced nutrient content, nutrient uptake and grain protein content of wheat under semi-arid conditions. Five irrigations (I_3) recorded

the highest N, P and K contents and uptake in grain and straw, along with higher grain protein content. Among nitrogen schedules, N_3 (33.33% N each at basal, CRI and jointing stages) recorded the highest N, P and K contents and uptake, as well as grain protein content. The irrigation $\times$ nitrogen interaction was non-significant for all parameters studied. Under the conditions evaluated in the present study, I_3 and N_3 emerged as the superior irrigation regime and nitrogen scheduling treatment, respectively, for improving nutrient uptake and grain nutritional quality of wheat.

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Disclaimers

The views and conclusions expressed in this article are solely those of the authors and do not necessarily represent the views of their affiliated institutions.

Conflict of interest

No conflicts of interest regarding the publication of this article.

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